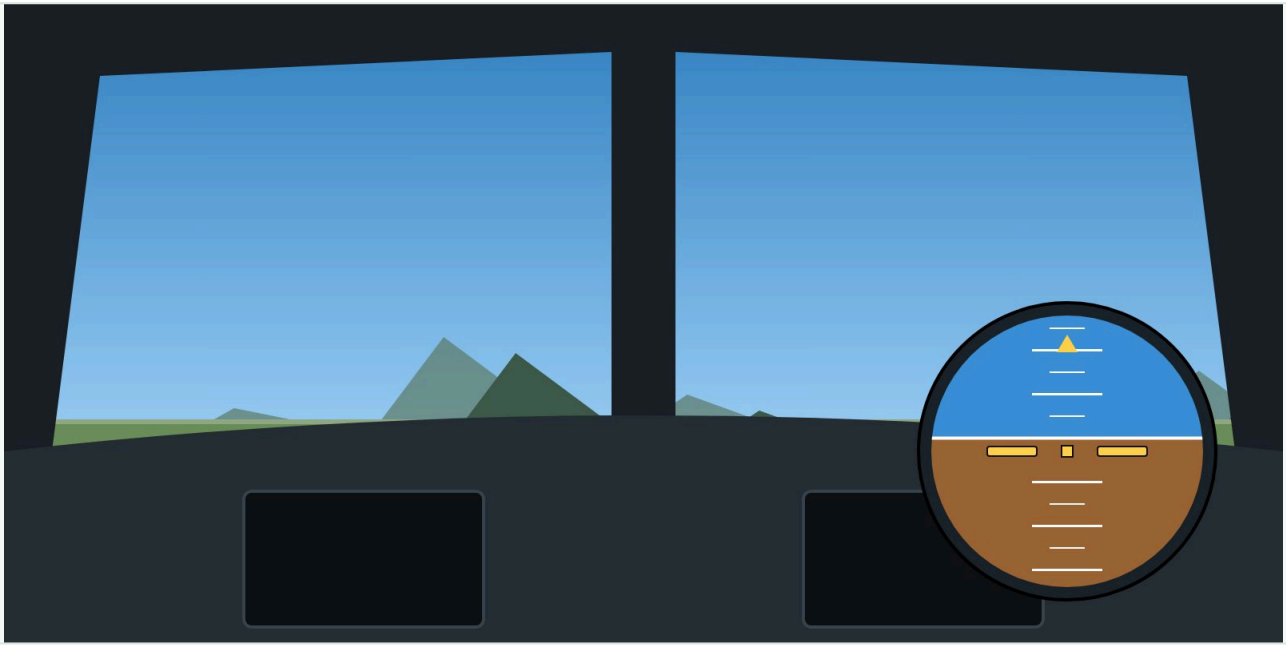
MODULE 1 · CFIT AWARENESS AND PREVENTION

Lesson 02: Altimeters, Safe Altitudes and Charts

Lesson 2 of 7 · about 7 min

LEARNING OBJECTIVES

- Apply correct altimeter setting procedures.
- Use minimum safe altitudes and chart information.
- Recognise when temperature corrections are needed.



Know the altimeter setting, the safe altitude and the terrain around you.

Altimeters

- Know the **units** used in the area: hectopascals or inches of mercury, feet or metres. Be ready to convert feet and metres.
- Be vigilant during radio transmissions: read back the altimeter setting in full, and verify if in doubt — QNH 1013 and 29.92 can be confused.
- Apply the correct setting for the **phase of flight** (transition altitude and level), with a cross-check and read-back procedure on the flight deck.
- **Cross-check the radio altimeter with the barometric altimeter**, especially in the approach.
- Fly higher than the minimum altitudes during atmospheric anomalies — strong winds over terrain, very low temperatures.
- Some EGPWS installations use **geometric altitude** (from GPS), which reduces errors from wrong altimeter settings or non-standard temperatures.

Safe altitudes

- Make sure the right charts are on board and **up to date**. Use colour terrain contours when available.

- Study the minimum safe altitude information — MSA, MORA / grid MORA, minimum sector altitudes and minimum vectoring altitudes.
- Know and fly at or above the safe altitude for your area of operation.
- **Descend below the MSA only** when established on a published instrument procedure, under radar control with an appropriate clearance, or when visual with the terrain.
- Know the height of the highest terrain or obstacle in the area and your position relative to it.

Cold temperatures

In air colder than ISA, the aircraft is **lower than the altimeter shows**. Apply cold temperature corrections to minimum altitudes according to PANS-OPS and the company procedure — see Lesson 24 of the Cold Weather Operations course.

KEY TAKEAWAYS

- Know the units; read back and cross-check every altimeter setting.
- Cross-check radio and barometric altimeters.
- Use up-to-date charts; stay at or above MSA unless on a published procedure, under radar control or visual.
- In cold air you are lower than indicated — apply corrections.

KNOWLEDGE CHECK

When may you descend below the MSA?

A. Whenever ATC is in contact

B. When established on a published procedure, under radar control with clearance, or visual with terrain

C. When the radio altimeter shows more than 1,000 ft

D. When the GPS is serviceable

Answer: B. Descent below MSA is only safe on a published procedure, under radar control with clearance, or when visual with the terrain.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.